LECTURE 1.3 · SHIP AND OCEAN SYSTEMS DESIGN

The Design Process

Papanikolaou (2026) §1.3.1–1.3.4; Table 1.5 · Session W1.2, second half · about 35 min

Bring: 0.6 The Reference Ship, slides 2 and 3

Tiago Santos, Luís Batista

Instituto Superior Técnico / University of Lisbon

tiago.santos@centec.tecnico.ulisboa.pt

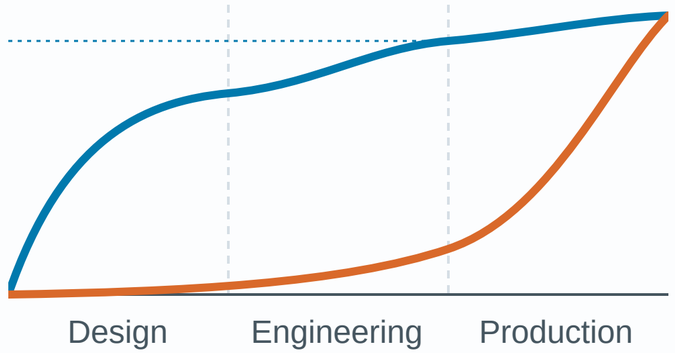
luis@veranavis.com

calendly.com/vnsd-luisbatista/meetme

The Life of a Ship: from Concept to Recycling

Stage	What happens
1 **Pre-contract**	Brief, yard offers, negotiation
2 **Design and engineering**	Basic design approved; detail and production drawings
3 **Production**	Steel cut, blocks built, outfitted, joined in the dock
4 **Launching**	Hull afloat; a payment milestone; outfitting goes on
5 **Commissioning and delivery**	Quay tests, sea trials, acceptance against the contract
6 **Operation and maintenance**	20 to 30 years: surveys, dry-docks, repairs, conversions
7 **Recycling**	Sold to a cash buyer, dismantled, steel recovered

Fig. 1 · Cost definition



Redrawn after Fischer and Holbach (2011), Fig. 1

Cost is assigned in design, but spent in production: by then, about 90 % is already fixed.

“The really big design decisions are made early in the design, hence this is the most important phase.” Andrews (2021)

The Actors

Owner side

Owner	Orders and owns
Manager	Operates, maintains
Charterer	Hires the ship
Cargo owner	Provides the cargo
Site team	Supervises the build

Finance and legal

Banks	Finance, guarantees
Export credit	State-backed loans
Insurers	Cover the risks
Broker	Order and sale deals
Lawyers	Draft the contract

Crew

Master and crew	Sail and maintain
Crewing agency	Recruits, trains crew

Design

Consultant	Concept, advice
Design office	Basic design
Detail design	Production drawings
Model basin	Tank tests

Build

Shipyard	Builds and delivers
Sub-contractors	Blocks, systems
Repair yard	Docking, repairs

Suppliers and services

OEMs	Engines, equipment
OEM service	Spares, after-sales
Steel suppliers	Plates and profiles
Bunker suppliers	Supply the fuel
Salvors	Emergency response

Rules and control

IMO	Sets conventions
Regional bodies	EU, coastal rules
Flag state / RO	Certificates
Port state	Inspects in port
Class approval	Approves drawings
Class survey	Surveys the build
Class in service	Periodical surveys

Ports

Port authority	Runs the port
Pilots and tugs	Bring the ship in
Terminals	Load and discharge
Ship agent	Handles port calls

End of life

Cash buyer	Buys for recycling
Recycling yard	Dismantles the ship

Sequential Engineering: the Design Spiral

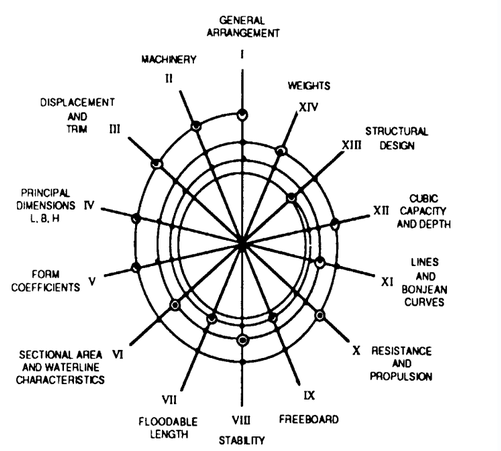
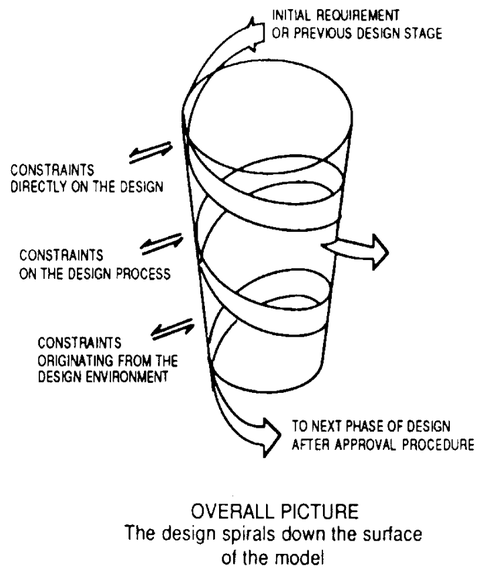


Fig. 3 Evan's general design diagram [40]

Evans (1959): the design spiral



OVERALL PICTURE
The design spirals down the surface of the model

Andrews (1981): the spiral in time, with outside constraints

1 One turn of the spiral

Choose L, B, T, D; then form, power and weights; check displacement, stability and cost.

2 Then go round again

Each turn adds detail and removes freedom, until the design balances: weight equals displacement.

3 Two additions

Buxton (1972): cost checked on every turn. Andrews (1981): time as a third dimension, and outside constraints.

4 Strength and weakness

Simple and traceable, even by hand; but slow, and a problem found late forces another full turn.

The spiral is **sequential**: one aspect at a time. Next, four ways to organise the turns.

Four Ways to Organise the Design Process

Approach	How it works	Gains	Costs	Where
1 Sequential (spiral)	One aspect at a time: dimensions, form, power, weights, stability, cost; repeated until the numbers agree	Simple, traceable, works by hand	Slow; a late problem forces a new turn	Classic method; phases A–D
2 Concurrent engineering	Design, production planning and purchasing at the same time, by one team sharing one model	Production problems caught early; shorter lead time	Early decisions; tight coordination	Shipyards today
3 Set-based design	Several options kept open in each discipline; those that clash with others are dropped as knowledge grows	No bad choice locked in early	More work up front; harder to manage	Naval ships
4 Design synthesis	One model computes mission, function, form, performance and economics for hundreds of variants; optimisation picks the best	Whole design space explored; best compromise	Needs a validated model	HOLISHIP; your D3 Ship Model

> All four still iterate. They differ in how many options stay open and how much runs in parallel.

Main Stages of Ship Design

I. Concept, Basic and Detail	Concept	Basic	Detail		
II. Classical four stages *	Concept	Preliminary	Contract	Detail	
III. Five stages	Concept	Preliminary	Contract	Detail	Production

Common to all definitions

Mission Requirements	**Basic Design Plans & Calculations for Classification Society**	**Statutory Design Plans & Calculations for Flag**	**Detail Design for Production**

The level of design depends on

Type of ship commercial or military, for example	**Client** owner, shipyard or another designer	**Shipyard** technological processes and documentation requirements	**Classification society** class notation
Manpower expertise skills and experience of the Production team	**Complexity** size, systems and novelty of the ship	**Integrated Logistic Support** spares, manuals and training	**Design workflow process** tools, data exchange and approvals

* Used in the textbook, Papanikolaou (2026) §1.3.2: A Concept, B Preliminary, C Contract, D Detailed; A + B = Basic Design

Different Perspectives on Design

1. By Scope / Level of Detail	Concept Design	Basic Design	Detail Design	As-Built / Construction Design		
2. By Design Phase / Project Stage	Feasibility Study	Preliminary Design	Functional Design	Production design	Delivery & Commissioning	
3. By Process Flow	Front-End Design (FED)	Engineering Design	Production Design	Integration & Verification	Support & Lifecycle	
4. By Discipline / Specialisation	Naval Architecture	Marine Engineering	Structural Engineering	Electrical & Control Systems	HVAC / Utilities	Outfitting
5. By Type of Vessel	Commercial	Defence	Specialised	High-Speed / Advanced	Green / Alternative	
6. By Client / Stakeholder	Shipowner / Operator	Shipyard	Classification Society	Flag State / Regulatory Bodies	Design Office / Engineering Firm	
7. By Tools / Technology	Traditional CAD	3D Modelling	CAE / FEA	CFD	PLM	Digital Twin
8. By Organisational Approach	In-house / Full Design Office	Outsourced / Partnered	Consortium / Joint Venture	Alliance / New Company		

Example of application

1 · Concept

Ship type · Deadweight · Propulsion system · Service speed

2 · Preliminary

General arrangement sketch · Preliminary lines plan · Propulsive power · Lightship weight · Initial cost

3 · Contract

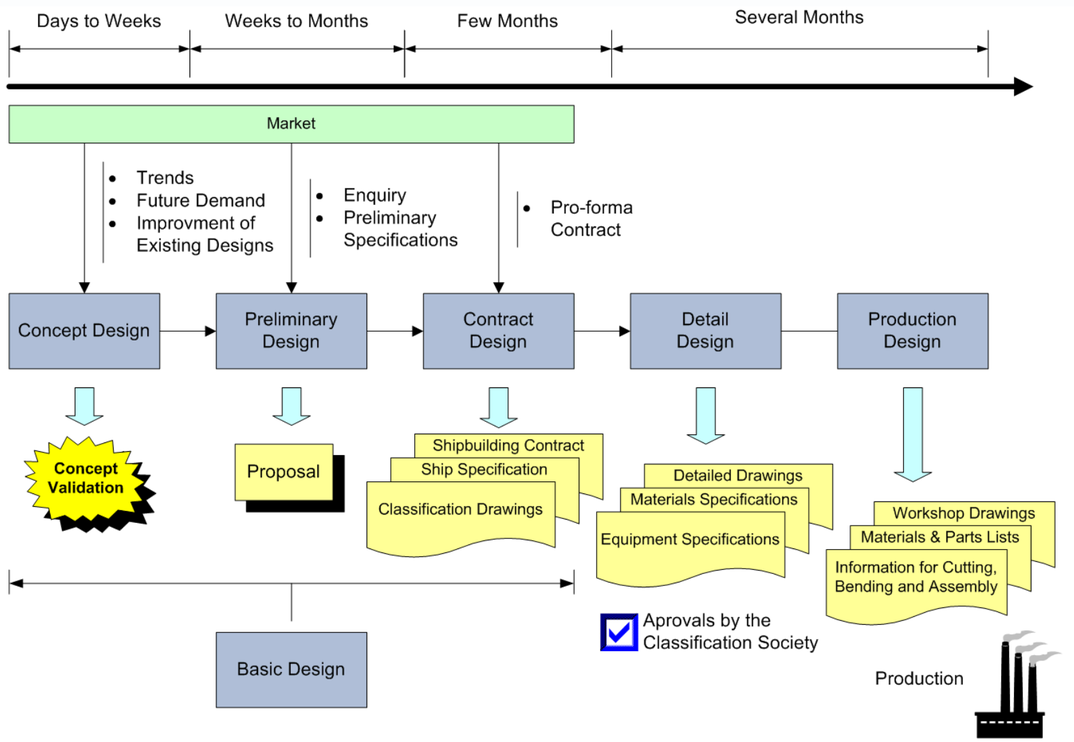
Specification and general arrangement · Lines plan for tank tests · Stability and structural computations · Some classification drawings · Single-line piping diagrams

4 · Detail

Faired lines or 3D hull model · Detailed general arrangement · All classification drawings · Building blocks · Construction planning · Piping and duct arrangements

5 · Production

Production drawings, to block level · Material and parts lists · Cutting, bending and assembly data · Piping isometrics



The stages along the time scale

Four Phases, Each a Turn of the Spiral

BASIC DESIGN · THIS COURSE · D1–D10

A · Concept

1st loop · feasibility

Starts from the owner’s brief. Ship type, deadweight, propulsion, speed; a first L, B, D, T, CB and PB; alternatives explored.

Today: about a day

B · Preliminary

2nd to 4th loop · optimum

Main dimensions, hull form, machinery, arrangement, structure, stability, freeboard, tonnage. GA sketch, power, lightship, cost.

Effort: about 15 × A

C · Contract

5th loop · specification

Faired lines and tank tests, structural and stability computations. The specification (machinery, class) and general arrangement signed with the contract.

Effort: about 17 × B

D · Detailed

Production

Detail and production (slide 8, stages 4 and 5): 3D model, class drawings, every detail, parts lists and workshop data.

Effort: the largest by far

After A + B the yard can price the ship and tender. These early decisions fix most of the cost of the ship’s **whole life**, from building to recycling.

The Contract Plans, Filed by SFI

No.	Plan	SFI groups	Starts
1	Outboard profile, GA	20, 24–26, 102	A
2	Inboard profile, GA	20–22, 24, 25, 102	A
3	GA, all decks and holds	20, 22, 23, 25, 30, 31, 35, 102	A
4	Crew quarters	51, 52, 54, 57, 58	C
5	Commissary spaces	55, 56	C
6	Lines	12, 20, 101, 102	A
7	Midship section	20, 23, 102	B
8	Steel scantling plan	20–25, 102	C
9	Machinery: plan views	22, 60, 63–66, 70–74	C
10	Machinery: elevations	22, 60, 63–66, 74	C
11	Machinery: sections	22, 60, 64–66, 70, 72	C
12	Main shafting	60, 63	C

No.	Plan	SFI groups	Starts
13	Power one-line diagram	85–89	C
14	Fire control diagram	50, 51, 81	C
15	Ventilation, AC diagram	37, 57	C
16	Piping system diagrams	35, 58, 70–73, 75, 80–82, 84	C
17	Heat balance, steam flow	64, 75, 84	C
18	Electric load analysis	65, 66, 86	C
19	Capacity plan	35, 70, 80, 102	B
20	Curves of form	101	B
21	Floodable length curves	101	C
22	Trim, stability booklet	15, 101	B
23	Damage stability	101	C

SFI main groups: 1 Ship general · 2 Hull · 3 Equipment for cargo · 4 Ship equipment · 5 Crew and passengers · 6 Machinery main components · 7 Systems for machinery · 8 Ship common systems. **Starts:** the phase where the plan first appears; shaded rows begin in A or B.

Four Things to Take Away

Design is short but decisive. Months of a 25 to 30 year life, yet it fixes about 90 % of what the ship will cost.

Design is a spiral. Each turn adds detail and removes freedom; concurrent, set-based and synthesis methods organise the turns, but the loop remains.

Four phases, A to D. Concept, Preliminary, Contract, Detailed. This course does A and B: basic design.

Each phase leaves documents. The contract plans of phase C, filed by SFI, start as sketches in A and B.

Next, Fri 6 Nov (W1.3): design methods, first displacement and main dimensions. Read §1.3.6–1.3.7 and §2.1–2.2. Bring a laptop.